

Borel parameters give countable Rosenthal index

Partial solution packet for Problem 7.10 of arXiv:1910.07273

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Answer to the first clause. For every Borel map $\theta : (0, 1) \rightarrow 2$, the trace space $C_Q(L^\theta)$ is Borel. Consequently $\text{ri}(L^\theta) < \omega_1$. The proof is qualitative and does not yet give the effective estimate in terms of the Borel class of θ requested in the second clause.

Source problem

Cabello Sánchez, Castillo, Marciszewski, Plebanek, and Salguero-Alarcón construct a compactification L^θ of

$$Q = \mathbb{Q} \cap (0, 1)$$

whose remainder is the double-arrow space. For each $x \in (0, 1)$ they fix a nondecreasing sequence $S_x \subset Q$ converging to x from the left and define

$$R_x^\theta = \begin{cases} P_x := Q \cap (0, x], & \theta(x) = 0, \\ P_x \setminus S_x, & \theta(x) = 1. \end{cases}$$

The Stone space of the Boolean algebra generated by the R_x^θ and the finite subsets of Q is L^θ . If D is countable dense in a separable Rosenthal compactum K , the index $\text{ri}(K)$ is countable exactly when the trace space $C_D(K) \subset \mathbb{R}^D$ is Borel.

The source proves that L^θ is Rosenthal and $C_Q(L^\theta)$ is analytic whenever θ is Borel, and asks the following on page 16 [1].

It is likely that twisted sums of c_0 and $C(\mathbb{S})$ that are of the form $C(K)$ with K Rosenthal compact can be examined more closely using the Rosenthal index. However, the following problem is open to us.

Problem 7.10. *Let $\theta : (0, 1) \rightarrow 2$ be Borel. Is $\text{ri}(L^\theta) < \omega_1$? Can we find an effective estimate of $\text{ri}(L^\theta)$ using the class of θ ?*

Proof intuition

The missing half of Borelness is coanalyticity. A function $f : Q \rightarrow \mathbb{R}$ fails to extend continuously to L^θ precisely when the closure of its graph has either two values over one point or an infinite value. Since L^θ is Rosenthal, graph-closure failures have sequential witnesses.

The special construction makes those witnesses Borel-codeable. A rational sequence converging to a split point y is governed by ordinary one-sided convergence, except that when $\theta(y) = 1$ the designated sequence S_y is rerouted from the upper side to the lower side. This gives an explicit Borel convergence relation. Hence nonextendibility is analytic, extendibility is coanalytic, and the source's analyticity result completes the argument by Souslin's theorem.

The Borel convergence relation

Write the remainder of L^θ as

$$\{y^- : 0 \leq y < 1\} \cup \{y^+ : 0 < y \leq 1\}.$$

The source's chosen binary-expansion model has the property that

$$E := \{(y, q) \in (0, 1) \times Q : q \in S_y\} \quad (1)$$

is Borel. Indeed, $y \mapsto S_y \in 2^Q$ is continuous off the countable set Q and therefore Borel everywhere.

Call a sequence $s = (q_n) \in Q^\mathbb{N}$ *free* if every fixed rational occurs only finitely often. For $0 < y < 1$, direct inspection of the generators R_x^θ gives

$$q_n \longrightarrow y^+ \text{ in } L^\theta \iff q_n \longrightarrow y \text{ in } \mathbb{R}, \text{ } s \text{ is free, and eventually } q_n < y \text{ and } (\theta(y) = 0 \text{ or } q_n \notin S_y), \quad (2)$$

$$q_n \longrightarrow y^- \text{ in } L^\theta \iff q_n \longrightarrow y \text{ in } \mathbb{R}, \text{ } s \text{ is free, and eventually } q_n > y \text{ or } (\theta(y) = 1 \text{ and } q_n \in S_y). \quad (3)$$

At the endpoints, $q_n \rightarrow 0^-$ (respectively 1^+) exactly when it is free and converges to 0 (respectively 1) in $\mathbb{R}$. A sequence converges to an isolated $q \in Q$ exactly when it is eventually constant at q .

To justify (2)–(3), convergence to the free ultrafilter y^+ means eventual membership in R_x^θ exactly for $x \geq y$, while convergence to y^- means eventual membership exactly for $x > y$. Real convergence and freeness automatically settle every generator with $x \neq y$: if $x > y$, only finitely many points of S_x lie below a fixed midpoint between y and x . The generator at $x = y$ gives precisely the displayed alternatives. Stabilization on the generators and finite sets is equivalent to stabilization on their Boolean algebra.

Because θ and the relation (1) are Borel, every condition in these formulas is Borel. Thus

$$\text{Conv}_\theta = \{(s, z) \in Q^\mathbb{N} \times L^\theta : s \text{ converges to } z\} \quad (4)$$

is Borel when L^θ is equipped with its standard Borel coding $Q \sqcup ([0, 1) \times \{-\}) \sqcup ((0, 1] \times \{+\})$.

Graph-closure criterion

Let $\psi : \mathbb{R} \rightarrow (-1, 1)$ be the homeomorphism

$$\psi(t) = \frac{t}{1 + |t|}.$$

For $f \in \mathbb{R}^Q$ put

$$A_f = \overline{\{(q, \psi(f(q))) : q \in Q\}}^{L^\theta \times [-1, 1]}.$$

Lemma 1. *The function f belongs to $C_Q(L^\theta)$ if and only if every vertical fiber of A_f is a singleton contained in $(-1, 1)$.*

Proof. If f extends continuously, the graph of the compactified extension is closed and equals A_f . Conversely, suppose every fiber is a singleton in $(-1, 1)$. The first-coordinate projection $A_f \rightarrow L^\theta$ is then a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. Thus A_f is the graph of a continuous map $g : L^\theta \rightarrow [-1, 1]$. Its compact image avoids $\{-1, 1\}$, so $\psi^{-1} \circ g$ is a real-valued continuous extension of f . $\square$

The product $L^\theta \times [-1, 1]$ is again a Rosenthal compactum: combine a Baire-one representation of L^θ with a countable-coordinate representation of the compact metric factor. By the Bourgain–Fremlin–Talagrand theorem it is Fréchet–Urysohn [2]. Therefore every point of A_f is the limit of a sequence from the displayed graph.

It follows from Lemma 1 that $f \notin C_Q(L^\theta)$ precisely when one of the following has a sequential witness:

- (i) two sequences $s, t \in Q^{\mathbb{N}}$ converge to the same $z \in L^{\theta}$, while $\psi(f(s_n)) \rightarrow u$, $\psi(f(t_n)) \rightarrow v$, and $u \neq v$;
- (ii) one sequence $s \in Q^{\mathbb{N}}$ converges to some z , while $\psi(f(s_n)) \rightarrow 1$ or -1 .

In view of the Borel relation (4), the set of tuples witnessing (i) or (ii) is Borel in the relevant product of standard Borel spaces. Its projection onto f is analytic. Hence $\mathbb{R}^Q \setminus C_Q(L^{\theta})$ is analytic and $C_Q(L^{\theta})$ is coanalytic.

Theorem 1. *For every Borel $\theta : (0, 1) \rightarrow 2$, the set $C_Q(L^{\theta})$ is Borel in $\mathbb{R}^Q$. Consequently*

$$\text{ri}(L^{\theta}) < \omega_1.$$

Proof. The source proves that $C_Q(L^{\theta})$ is analytic for every Borel θ [1, Lemma 7.8]. The preceding graph argument shows that it is coanalytic. Souslin's theorem now implies that it is Borel. Since Q is countable and dense in L^{θ} , this is exactly the assertion that the Rosenthal index is countable. $\square$

Upgrade attempt and remaining clause

The argument was pushed one step further by writing the one-sided Cauchy conditions implicit in (2)–(3) and tracking the Borel class of θ . This does not yet produce a class-sensitive ordinal bound: failure of a Cauchy condition requires projection over the continuum of split points, and the final analytic/coanalytic separation theorem does not retain a usable Borel-rank estimate. Effective descriptive-set separation may yield a bound relative to a particular Borel code for θ , but no bound depending only on the Borel class is proved here.

Thus Theorem 1 completely answers the yes/no first clause of Problem 7.10, while its second, quantitative clause remains open.

Bounded novelty check

On 11 August 2026, the exact arXiv id, title, Problem 7.10 text, and close variants involving $\text{ri}(L^{\theta})$, Borel parameters, and twisted sums were checked in the run indexes and by bounded web/arXiv search. The searches returned the source paper and a later thesis restating the construction, but no later answer to Problem 7.10. This is evidence for novelty, not a definitive guarantee.

References

- [1] F. Cabello Sánchez, J. M. F. Castillo, W. Marciszewski, G. Plebanek, and A. Salguero-Alarcón, *Sailing over three problems of Koszmider*, arXiv:1910.07273, version 2, 2020.
- [2] J. Bourgain, D. H. Fremlin, and M. Talagrand, *Pointwise compact sets of Baire-measurable functions*, Amer. J. Math. **100** (1978), 845–886.
- [3] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.